

Intermediate Microeconomic

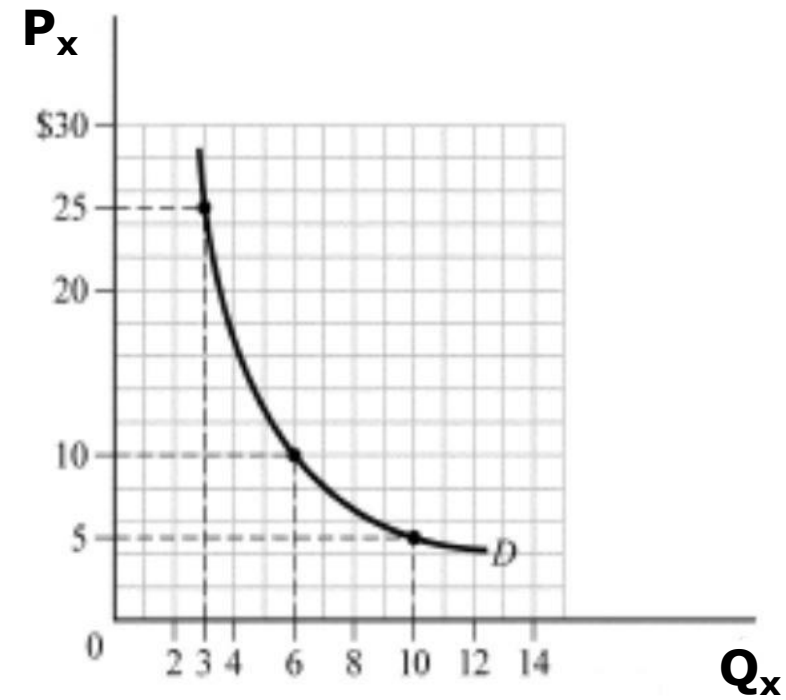
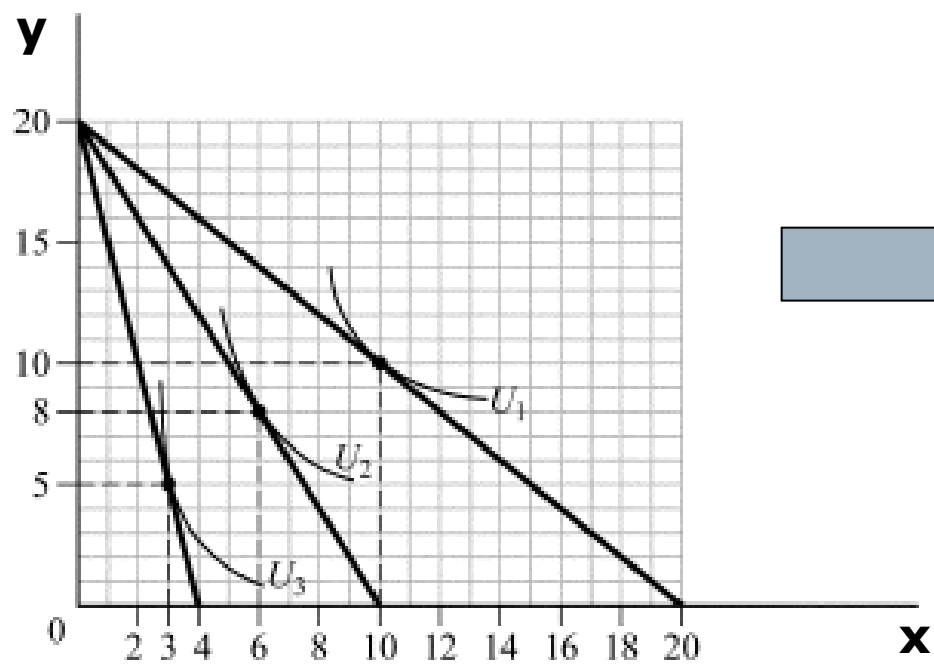
Part two: Choice and Demand

Week 3b: Demand

Yuanning Liang

Compensated Demand Curves

- The actual level of utility varies along the uncompensated (Marshallian) demand curve



Compensated Demand Curves

- The actual level of utility varies along the Marshallian demand curve
- As the price of x falls, the individual moves to higher indifference curves
 - it is assumed that nominal income is held constant as the demand curve is derived
 - this means that “real” income rises as the price of x falls
- An alternative approach holds “real” income (or utility) constant while examining reactions to changes in p_x
 - the effects of the price change are “compensated” so as to constrain the individual to remain on the same indifference curve
 - reactions to price changes include only substitution effects

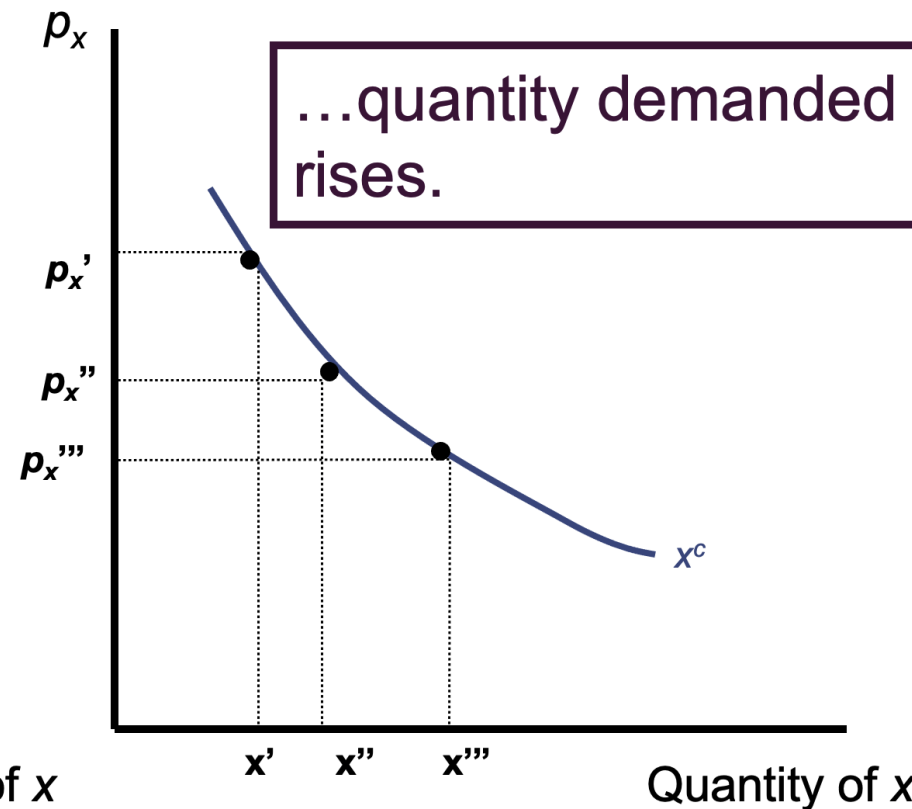
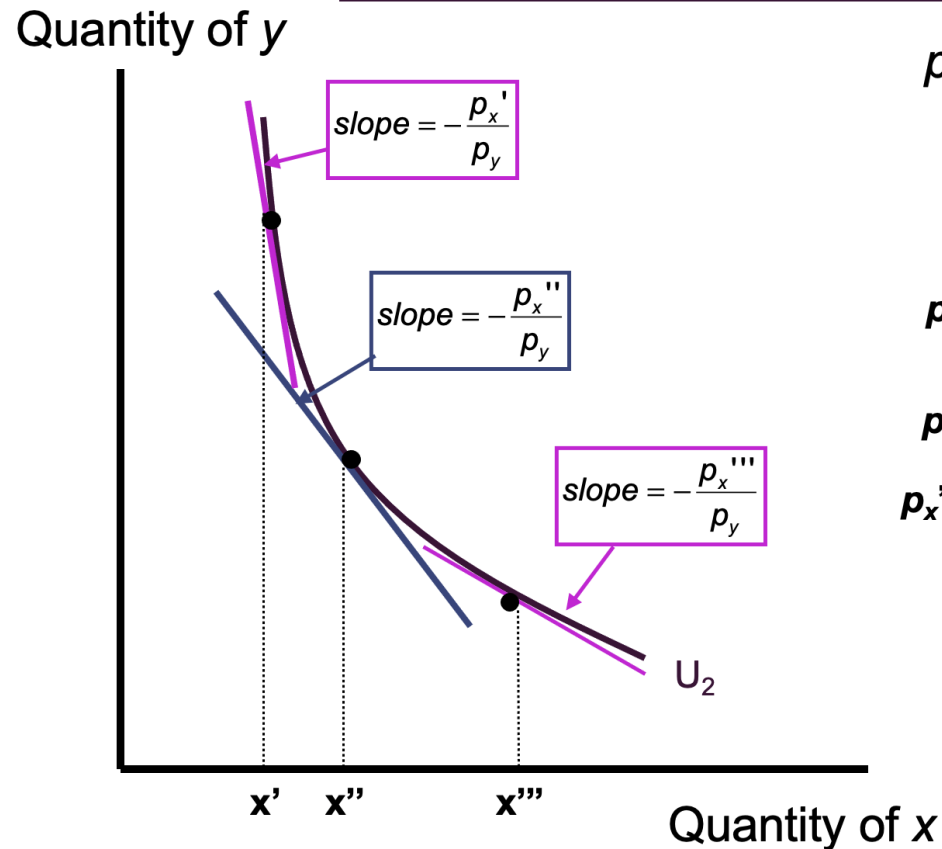
Compensated Demand Curves

- A compensated (Hicksian) demand curve shows the relationship between the price of a good and the quantity purchased assuming that other prices and utility are held constant.
- The compensated demand curve is a two-dimensional representation of the compensated demand function

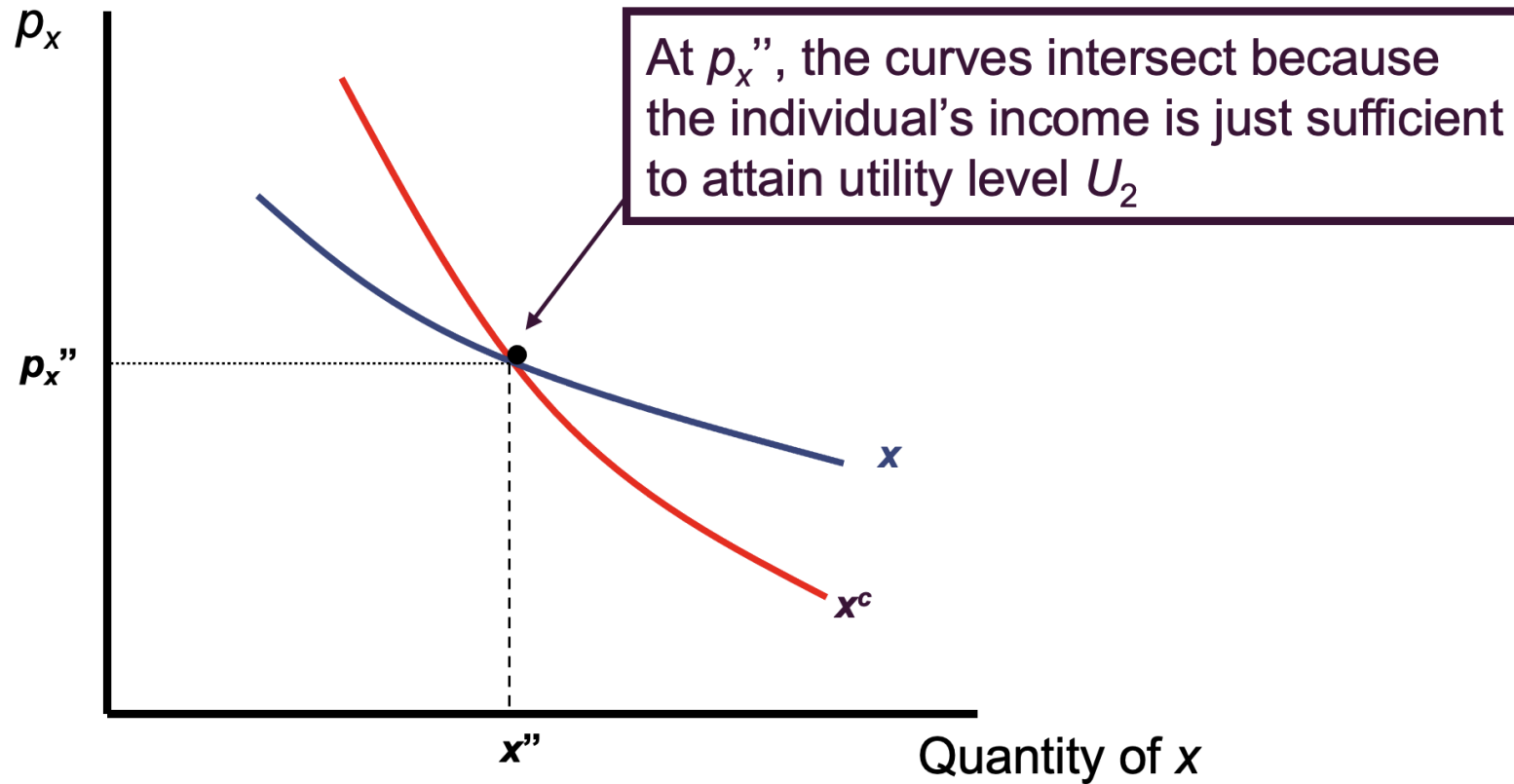
$$x^* = x^c(p_x, p_y, U)$$

Compensated Demand Curves

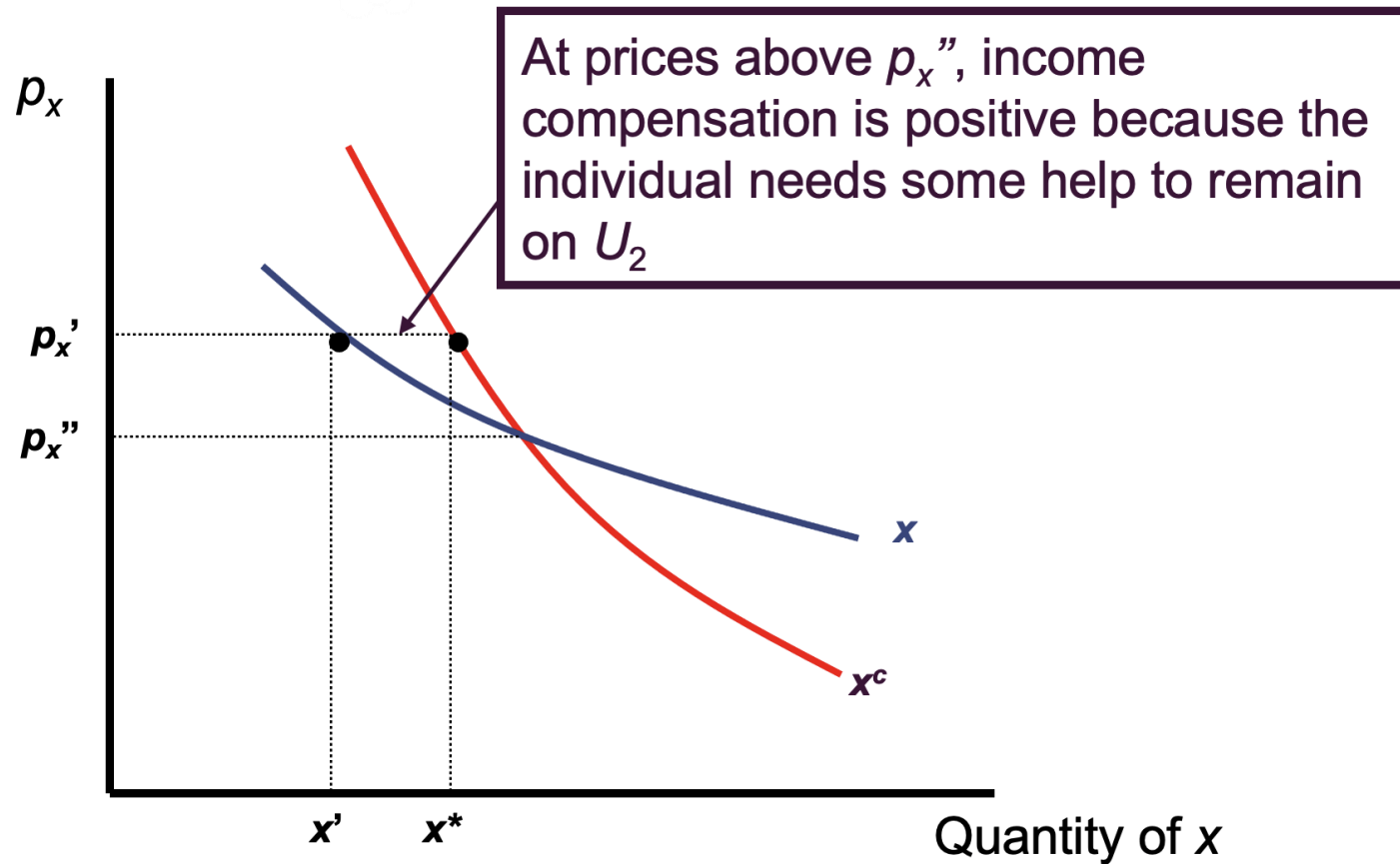
Holding utility constant, as price falls...



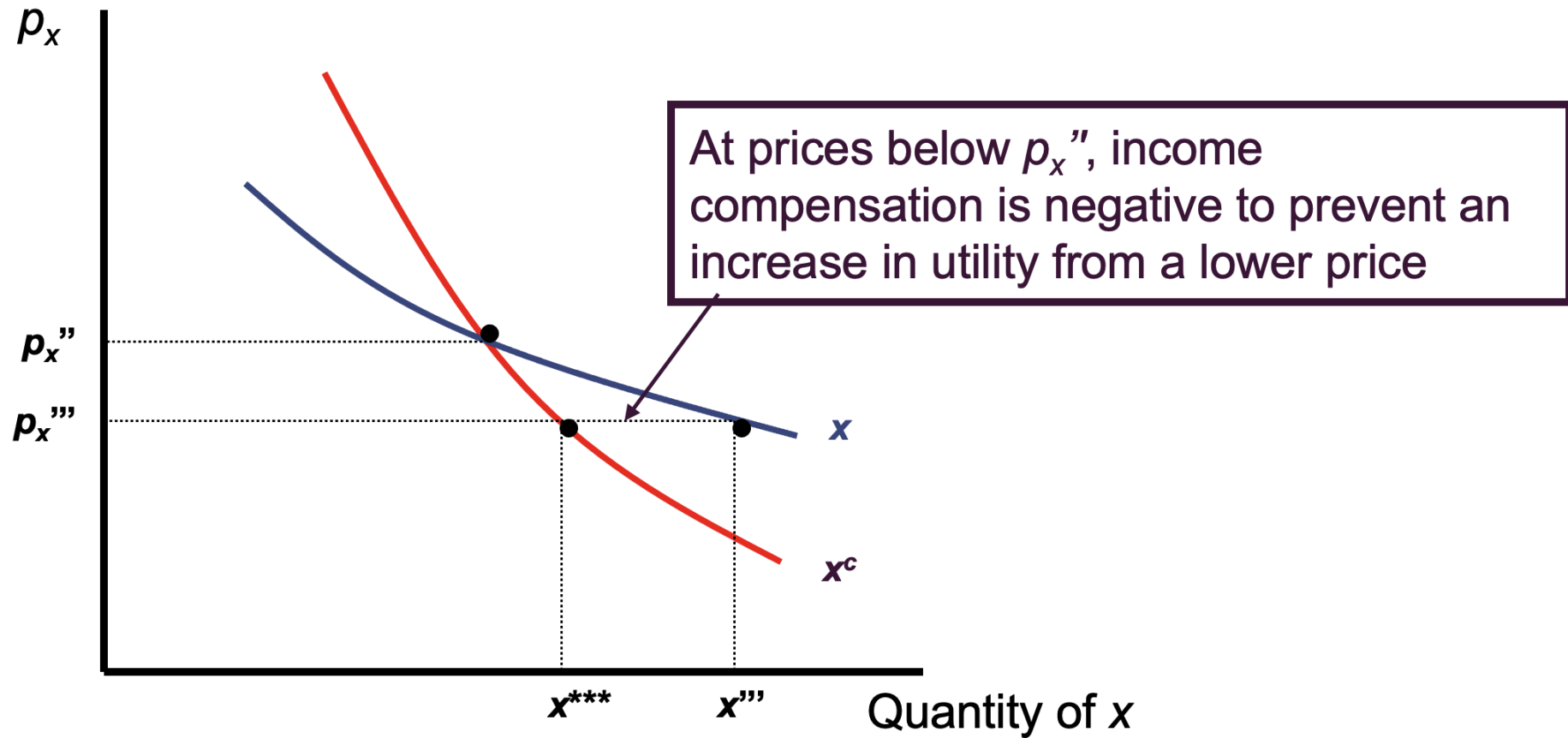
Compensated & Uncompensated Demand



Compensated & Uncompensated Demand



Compensated & Uncompensated Demand



Compensated & Uncompensated Demand

- For a normal good, the compensated demand curve is **less responsive** to price changes than is the uncompensated demand curve
 - the **uncompensated** demand curve reflects both income and substitution effects
 - the **compensated** demand curve reflects only substitution effects (because income is compensated)

Compensated Demand Functions

- Suppose that utility is given by

$$\text{utility} = U(x,y) = x^{0.5}y^{0.5}$$

- The uncompensated (Marshallian) demand functions are

$$x = I/2p_x$$

$$y = I/2p_y$$

- The indirect utility function is

$$\text{utility} = V(I, p_x, p_y) = \frac{I}{2p_x^{0.5}p_y^{0.5}}$$

Compensated Demand Functions

- To obtain the compensated demand functions, we can solve the indirect utility function for I and then substitute into the Marshallian demand functions

$$x^c = \frac{V p_y^{0.5}}{p_x^{0.5}} \qquad y^c = \frac{V p_x^{0.5}}{p_y^{0.5}}$$

- Demand now depends on utility (V) rather than income
- Increases in p_x reduce the amount of x demanded
 - only a substitution effect

Duality

- Remember the expenditure function

$$\text{minimum expenditure} = E(p_x, p_y, U)$$

- Then, by definition

$$x^c(p_x, p_y, U) = x[p_x, p_y, E(p_x, p_y, U)]$$

- The quantity demanded is equal for both Marshallian and Hicksian demand functions when income is exactly what is needed to attain the required utility level.
-

Deriving the Slutsky equation

- Slutsky discovered that changes to demand from a price change are always the sum of a pure substitution effect and an income effect.

- We can differentiate the compensated demand function

$$x^c(p_x, p_y, U) = x[p_x, p_y, E(p_x, p_y, U)]$$

- and get

$$\frac{\partial x^c}{\partial p_x} = \frac{\partial x}{\partial p_x} + \frac{\partial x}{\partial E} \cdot \frac{\partial E}{\partial p_x}$$

Deriving the Slutsky equation

$$\frac{\partial x}{\partial p_x} = \frac{\partial x^c}{\partial p_x} - \frac{\partial x}{\partial E} \cdot \frac{\partial E}{\partial p_x}$$

- The first term is the slope of the compensated demand curve
 - the substitution effect
- The second term measures the way in which changes in p_x affect the demand for x through changes in purchasing power
 - the income effect

A mathematical representation of income and substitution effects

The substitution effect can be written as

$$\text{substitution effect} = \frac{\partial x^c}{\partial p_x} = \frac{\partial x}{\partial p_x} \bigg|_{U=\text{constant}}$$

The income effect can be written as

$$\text{income effect} = -\frac{\partial x}{\partial E} \cdot \frac{\partial E}{\partial p_x} = -\frac{\partial x}{\partial I} \cdot \frac{\partial E}{\partial p_x}$$

The Slutsky Equation

- Note that $\partial E / \partial p_x = x$ (as $E = p_x x + p_y y$)
 - a \$1 increase in p_x raises necessary expenditures by x dollars

- The utility-maximization hypothesis shows that the substitution and income effects arising from a price change can be represented by

$$\frac{\partial x}{\partial p_x} = \text{substitution effect} + \text{income effect}$$

$$\frac{\partial x}{\partial p_x} = \left. \frac{\partial x}{\partial p_x} \right|_{U=\text{constant}} - x \frac{\partial x}{\partial I}$$

The Slutsky Equation

$$\frac{\partial x}{\partial p_x} = \underbrace{\frac{\partial x}{\partial p_x} \bigg|_{U=\text{constant}}}_{\text{substitution effect}} \underbrace{- x \frac{\partial x}{\partial I}}_{\text{income effect}}$$

- The first term is the **substitution effect**
 - always negative as long as *MRS* is diminishing
 - the slope of the compensated demand curve must be negative

- The second term is the **income effect**
 - if *x* is a normal good, then $\partial x / \partial I > 0$
 - the entire income effect is negative
 - if *x* is an inferior good, then $\partial x / \partial I < 0$
 - the entire income effect is positive

A Slutsky Decomposition

- We can demonstrate the decomposition of a price effect using the Cobb-Douglas example studied earlier

- The Marshallian demand function for good x is

$$x(p_x, p_y, I) = \frac{0.5I}{p_x}$$

- The Hicksian (compensated) demand function for good x is

$$x^c(p_x, p_y, V) = \frac{Vp_y^{0.5}}{p_x^{0.5}}$$

A Slutsky Decomposition

- The overall effect of a price change on the demand for x is

$$\frac{\partial x}{\partial p_x} = \frac{-0.5I}{p_x^2}$$

- This total effect is the sum of the two effects that Slutsky identified
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A Slutsky Decomposition

- The substitution effect is found by differentiating the compensated demand function

$$\text{substitution effect} = \frac{\partial x^c}{\partial p_x} = \frac{-0.5Vp_y^{0.5}}{p_x^{1.5}}$$

- We can substitute in for the indirect utility function ($V(p_x, p_y, I)$)

$$\text{substitution effect} = \frac{-0.5(0.5Ip_x^{-0.5}p_y^{-0.5})p_y^{0.5}}{p_x^{1.5}} = \frac{-0.25I}{p_x^2}$$

A Slutsky Decomposition

- Calculation of the income effect is easier

$$\text{income effect} = -x \frac{\partial x}{\partial I} = -\left[\frac{0.5I}{p_x} \right] \cdot \frac{0.5}{p_x} = -\frac{0.25I}{p_x^2}$$

Practice Example: Demand for nonmarket goods

- The theory of **revealed preference** was proposed by Paul Samuelson in the late 1940s
- The theory defines a principle of rationality based on observed behavior and then uses it to approximate an individual's utility function
- For example, it helps us to estimate demand for nonmarket goods.

Revealed Preference

- Consider two bundles of goods: **A** and **B**
- If the individual can afford to purchase either bundle but chooses **A**, we say that **A** had been revealed preferred to **B**
- Under any other price-income arrangement, **B** can never be revealed preferred to **A**

Applications of revealed preference

- There is no clear way to **directly** value changes in quantities of environmental/recreational goods.
 - Why?
-

Applications of revealed preference

- ❑ There is no clear way to **directly** value changes in quantities of environmental/recreational goods.
 - ❑ Why?
 - ❑ There aren't any markets for them!
 - ❑ Is there a way we can reveal the value of these goods?
-

Applications of revealed preference

- ❑ There is no market for orcas
- ❑ Suppose there's a massive decline in orcas off the Washington coast, what happens?
- ❑ We will likely see demand for sightseeing tours go down.
- ❑ This drops the price of tours.
- ❑ A non-market good had an effect on a market price.



Applications of revealed preference

- ❑ One way to circumvent this problem is to look at private goods that interact with the environmental good
 - ❑ If there are changes in the environmental good, holding everything else fixed, that should be reflected in *some way* in changes in the price of the related private good
 - ❑ This change in price can tell us something about how people value the change in the environmental good
 - ❑ 1. Hedonics method
 - ❑ 2. Travel cost model
-

2. Travel cost model

- ❑ Recreational areas **have value**
 - ❑ Their quality also has value
 - ❑ Not placing a value on recreation is essentially giving it a value of **zero**
 - ❑ This is likely inappropriate
 - ❑ If someone dumped toxic waste in Yellow River does that have zero cost?
-

2. Travel cost model

- ❑ The travel cost method uses observable data on recreation visitation to infer the recreational value of environmental amenities
 - ❑ The central idea is that the time and travel cost expenses that people incur to visit a site represent the **price** of access to the site
 - ❑ This means that people's willingness-to-pay to visit can be estimated based on the number of visits they make to sites of different prices
 - ❑ This gives us a **demand curve** for sites/amenities, so we can value changes in these environmental amenities
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Hotelling

- ❑ After WWII, the U.S. national park service solicited advice from economists on methods for quantifying the value of specific park properties
 - ❑ Would total entrance fee that people pay measure the value?
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Hotelling

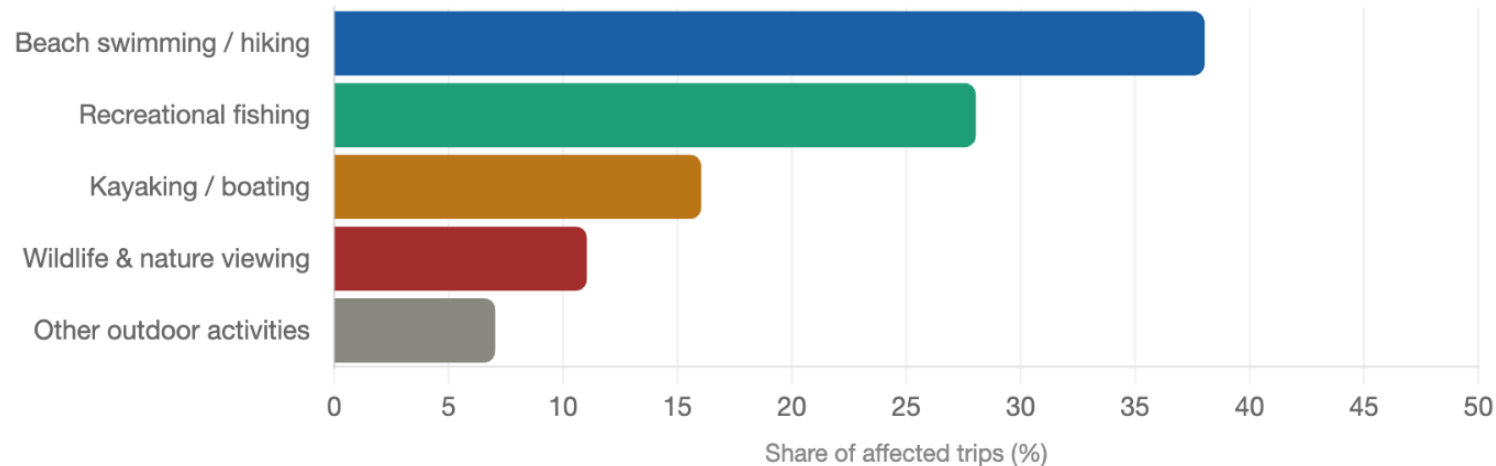
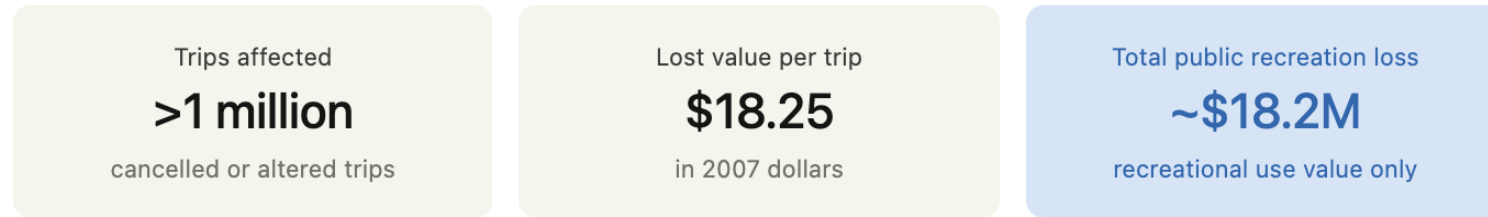
- ❑ After WWII, the U.S. national park service solicited advice from economists on methods for quantifying the value of specific park properties
 - ❑ Would total entrance fee that people pay measure the value?
 - ❑ **No!**
 - ❑ Harold Hotelling proposed the first indirect method for measuring the demand of a non-market good in 1947
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Cosco Busan



Chart 2 · 2007 Cosco Busan Oil Spill: TCM Loss Estimation

~540,000 gallons of fuel oil spilled into San Francisco Bay → beaches and fishing areas closed



Source: NOAA Office of Response & Restoration · California Recreational Fisheries Survey

How the Travel Cost Method works here

Someone living near the bay drives 10 min to the beach — low travel cost, visits often.

Someone from San Jose drives an hour — higher cost, visits less frequently.

After the spill, **both groups reduced their trips**.

Economists estimated how many trips *would have* occurred without the spill, then multiplied by the lost value per trip — the **lost consumer surplus**. That's the TCM in action.

Theoretical foundation

- Here's the theory for how we can use observed data to tell us something about willingness to pay
 - Consider a single consumer and a single recreation site
 - The consumer has:
 - Total number of recreation trips: x , to site of quality: q
 - Total budget of time: T
 - Working time: H
 - Non-recreation, non-work time: I
 - Time to recreation site: t
 - Hourly wage: w
 - Money cost of reaching the site: c
 - consumption: z
-

Theoretical foundation

- This lets us write down the consumer's utility maximization problem:

$$\max_{x,z,l} U(x, z, l, q) \quad \text{subject to:} \quad \underbrace{wH = cx + z}_{\text{money budget}}, \quad \underbrace{T = H + L + tx}_{\text{time budget}}$$

- Combine the two constraints to get:

$$\max_{x,z,l} U(x, z, l, q) \quad \text{subject to:} \quad \underbrace{wT = z + (c + wt)x + wl}_{\text{combined money/time budget}}$$

Theoretical foundation

- Let $Y = wT$ be the consumer's *full income*, their money value of total time budget
- Let $p = c + wt$ be the consumer's *full price*, their total cost to reach the site
- Then we can write the problem as:

$$\max_{x,z,l} U(x, z, l, q) \quad \text{subject to: } \underbrace{Y = z + px + wl}_{\text{combined budget}}$$

- Solve the constraint for z and substitute into the utility function...
-

Theoretical foundation

$$\max_{x,l} U(x, Y - px - wl, l, q)$$

This has first-order conditions:

$$[x] \quad U_x - pU_z = 0 \rightarrow \frac{U_x}{U_z} = p$$

and

$$[l] \quad -wU_z + U_l = 0 \rightarrow \frac{U_l}{U_z} = w$$

Theoretical foundation

- $\frac{U_x}{U_z} = p$ tells us the consumer equates the marginal rate of substitution between recreational trips and consumption to be the full price of the recreational trip
 - What does this mean?
 - The value of the recreational trip to the consumer, in dollar terms, is revealed by the full price p
-

Theoretical foundation

$$U_x - pU_z = 0 \quad - wU_z + U_l = 0$$

- The above FOCs are two equations, the consumer had two choices (x,l) so we had two unknowns
 - We can thus solve for x (and l) as a function of the parameters (p,Y,q):
$$x = f(p, Y, q)$$
 - This is simply the consumer's **demand curves** for recreation as a function of the full price p, full budget Y, and quality q
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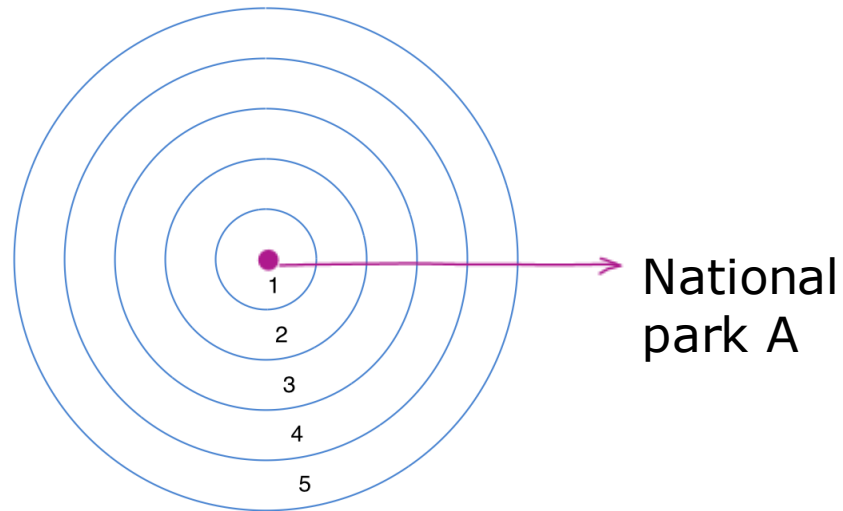
Theoretical foundation

$$x = f(p, Y, q)$$

- If we observe consumers going to sites of different full prices $p_1, p_2, \dots, p_n$, we are moving up and down their recreation demand curve
 - This lets us trace out the demand curve
 - Changing Y or q shifts the demand curve in or out: these are income and quasi-price effects
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Zonal (single-site) model

- Here's the most basic travel cost model to start:
- A simple example of zones 1-5



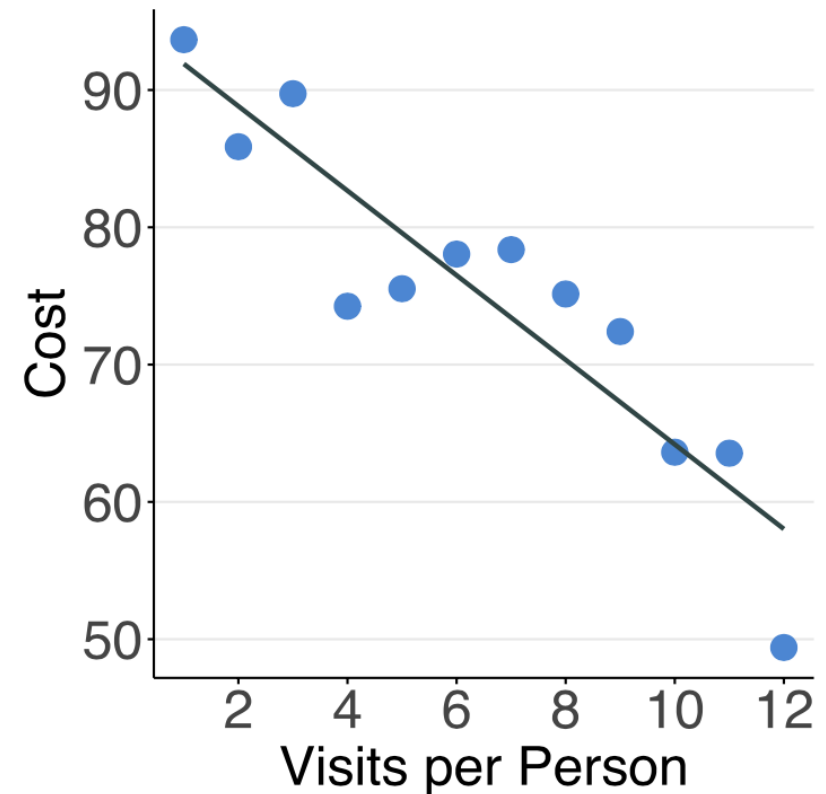
Zonal (single-site) model

Demand curve is simply from estimating

Visits per person = $f[\text{travel cost per visit, entrance fee, (and travel costs to other sites), relevant socioeconomic data (income and education, etc)}] + \epsilon$

Or even more simply,

$$\text{Visits per person} = \beta_0 + \beta_1 \text{travel cost} + \epsilon$$



Issues and concerns with the single-site model?

- ❑ It ignores non-use value (automatically zero for non-users)
 - ❑ What are the right zones to choose?
 - ❑ What is the right functional form for demand?
 - ❑ How do we measure the opportunity cost of time?
 - ❑ How do we treat multi-purpose trips?
 - ❑ How do we value particular site attributes? Can't disentangle them at a single site
-